

Hey, everybody. Welcome to your chapter 19 review. We're going to spend some time on SAQ, which is short for Speed, Agility, and Quickness. This is another important piece to the integrated training puzzle.

Speed is the velocity of movement in a given direction. Agility is the ability to sprint in multiple directions or what we call running and cutting, while still maintaining postural control. And quickness refers to the ability to react to a stimulus and appropriately change the motion of the body in response to that stimulus.

Picture a group of kids playing tag. What do you observe? Speed? Yep. Agility? Absolutely. Quickness? You better hope so. These are qualities that define youthful movement.

The good news is there are ways to preserve and improve those qualities in any person, regardless of age, through fitness training. SAQ training stimulates muscular, neurological, connective tissue, and cardiovascular fitness adaptations by presenting the body with scenarios requiring coordinated movements, reaction to stimuli, and quick bursts of speed. These challenges can promote improvements in physical performance and movement, including the ability to accelerate, decelerate, and dynamically stabilize the entire body.

By the end of your studies on this chapter, you should be able to describe speed, agility, and quickness training and its purpose, discuss the importance of SAQ training for a variety of populations, design an SAQ training program for clients at any level of training, and perform, describe, and instruct various SAQ training exercises.

Typically and similar to plyometrics, SAQ training is associated with a more athletic population. However, it can play a vital role in all training programs. When implemented correctly, SAQ training can add variety to client programs and lead to valuable fitness improvements for many types of individuals, such as youth, older adults, or weight loss clients to name a few.

These exercises present the body and mind with various challenges that can promote tons of positive improvements in physical performance. The ability to react quickly, control the body through coordinated movement patterns, and change directions under control and with speed will enable clients to participate in recreational activities throughout life in a playful and safe manner.

As always, you'll get a handle on your client's goals, baseline fitness, and any concerns during your assessments. After ensuring that a client is ready, you can include SAQ drills in a progressive manner, beginning with low volume, simple movements, and then progress to faster, more complex movements at higher volumes. This progression may happen over a span of several months. SAQ drills should focus first and foremost on quality of movement. You want to be sure your client has mastered proper technique before adding new variables to increase the challenge.

Then the speed of movement can increase to maximize the response and performance of the neuromuscular system. Let's look at the three components of SAQ in greater detail. Speed is the product of stride rate and stride length. Stride rate is the number of strides taken in a given amount of time or distance, and stride length is the distance covered in one stride.

Form is important here too. Proper running mechanics will enable maximum force generation through biomechanical efficiency. And interestingly enough, certain aspects of speed are dependent on genetic factors. However, speed is a skill that can be learned and developed through an integrated training program. Not every client needs to run as fast as an elite athlete, but speed improvements can have both performance and health-related advantages for everyone.

Speed specific exercises, such as sprinting drills, stimulate muscle tissue and nerves in ways that other training methods don't. Agility necessitates the ability to accelerate, decelerate, stabilize, and change direction while maintaining postural control. It requires high levels of coordination to maintain one's center of gravity, while changing directions at various speeds.

But it's well worth it. Proper agility training can help reduce injury risk by enhancing the body's ability to effectively control eccentric forces in all planes of motion, as well as improving the structural integrity of connective tissue. It's important to note, however, that agility and Change of Direction, or COD, are separate.

Agility involves reacting to a signal or stimulus and changing the body's sprinting direction in response. On the other hand, change of direction doesn't involve reacting to a signal, but still requires the person to accelerate, decelerate, and change directions. So more like an athlete running through a predetermined obstacle course.

This differentiation is important to understand as you design agility drills, which should involve a response to a signal versus only working on cod. Quickness is the ability to react and change body position with maximal rate of force production in all planes of motion and from all body positions during dynamic activities. Quickness involves assessing visual, auditory, or kinesthetic stimuli and provide the appropriate physical response as fast as possible, such as hitting a baseball, returning a tennis serve, or swerving to avoid a car accident.

When you're working through the difference between quickness and agility, here's a helpful way to look at it. Agility often requires a person to react to a stimulus and respond accordingly, such as a basketball player moving in various directions to avoid a defender. Quickness also requires a person to react to a stimulus. However, the response doesn't require running or sprinting.

A good example of this is a hockey goalkeeper defending against a penalty shot. When implemented correctly, multiple client types can benefit from SAQ training. For youth. It's an effective way of providing a variety of exposures to physiologic, neuromuscular, and biomechanical demands. This results in the further development of physical ability and a decrease in the likelihood of athletic injury.

For weight loss, SAQ exercises can contribute to an increase in caloric expenditure due to elevated heart rate and respiration rate. It's great to be able to utilize some SAQ drills to break up more traditional forms of cardio training as well. Older adults can achieve many of the same benefits as weight loss clients, while also experiencing a positive impact on bone density, power, reaction time, and coordination.

SAQ training adaptations are important to prevent falls and work toward the maintenance of independence later in life. Activities that may become more challenging with age, like golf, tennis, or skiing, can be enjoyed again with some training. And of course, your apparently healthy adult population and more athletically inclined clients will enjoy all the variation and the fun, fast paced nature of SAQ exercises.

The increased physical demands of SAQ training can improve athleticism, movement proficiency, and aid in injury prevention. Just be sure to do a thorough assessment of your client up front and follow a comprehensive warm up protocol. Just like anything else, integrating SAQ drills and strategies into a training program is going to come down to the individual.

Make sure it is specific to the individual you are working with, and make adjustments wherever you need to based on your client's unique needs and capabilities. Keep in mind to monitor all exercises for precise technique to minimize the risk of injury. SAQ training is fun, invigorating, and helps create, improve and enhance that youthful energy we're all trying to hold on to. With that, I'll leave you to it. Until next time, take care and happy studying.